

# Software Requirements Specification

Renewal & Upsell Advisor

Agent ID: S-007

Suite Category: Sales Suite

Version: 1.0

December 28, 2025

**Confidential**

# Contents

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| <b>1</b> | <b>Project Overview</b>                       | <b>5</b>  |
| 1.1      | Project Title . . . . .                       | 5         |
| 1.2      | Agent Identification . . . . .                | 5         |
| 1.3      | Executive Summary . . . . .                   | 5         |
| <b>2</b> | <b>Objectives</b>                             | <b>5</b>  |
| 2.1      | Primary Objectives . . . . .                  | 5         |
| 2.2      | Secondary Objectives . . . . .                | 6         |
| <b>3</b> | <b>Scope</b>                                  | <b>6</b>  |
| 3.1      | In-Scope . . . . .                            | 6         |
| 3.2      | Out-of-Scope . . . . .                        | 7         |
| <b>4</b> | <b>Business Scenario</b>                      | <b>7</b>  |
| 4.1      | Use Case Context . . . . .                    | 7         |
| 4.2      | Primary Business Scenario . . . . .           | 7         |
| <b>5</b> | <b>Milestones &amp; Timeline</b>              | <b>9</b>  |
| 5.1      | 7-Day Development Sprint . . . . .            | 9         |
| 5.2      | Detailed Task Breakdown . . . . .             | 9         |
| 5.3      | Critical Path Dependencies . . . . .          | 11        |
| <b>6</b> | <b>Roles &amp; Responsibilities</b>           | <b>11</b> |
| 6.1      | Technical Team Structure . . . . .            | 11        |
| 6.2      | Team Size . . . . .                           | 12        |
| <b>7</b> | <b>Deliverables</b>                           | <b>12</b> |
| 7.1      | Technical Deliverables . . . . .              | 12        |
| 7.2      | Documentation Deliverables . . . . .          | 14        |
| <b>8</b> | <b>Tech Stack</b>                             | <b>14</b> |
| 8.1      | Frontend Technologies . . . . .               | 14        |
| 8.2      | Backend Technologies . . . . .                | 15        |
| 8.3      | Machine Learning Stack . . . . .              | 15        |
| 8.4      | Database & Storage . . . . .                  | 15        |
| 8.5      | Integration & APIs . . . . .                  | 16        |
| 8.6      | DevOps & Infrastructure . . . . .             | 16        |
| 8.7      | Development Tools . . . . .                   | 16        |
| <b>9</b> | <b>Functional Requirements</b>                | <b>17</b> |
| 9.1      | FR-1: Contract & Renewal Tracking . . . . .   | 17        |
| 9.1.1    | FR-1.1: Contract Data Ingestion . . . . .     | 17        |
| 9.1.2    | FR-1.2: Renewal Timeline Monitoring . . . . . | 17        |
| 9.2      | FR-2: Churn Risk Prediction . . . . .         | 17        |
| 9.2.1    | FR-2.1: Risk Scoring Model . . . . .          | 17        |
| 9.2.2    | FR-2.2: Risk Segmentation . . . . .           | 17        |
| 9.2.3    | FR-2.3: Early Warning Alerts . . . . .        | 18        |

|           |                                                     |           |
|-----------|-----------------------------------------------------|-----------|
| 9.3       | FR-3: Upsell & Cross-sell Identification . . . . .  | 18        |
| 9.3.1     | FR-3.1: Expansion Opportunity Detection . . . . .   | 18        |
| 9.3.2     | FR-3.2: Cross-sell Recommendations . . . . .        | 18        |
| 9.3.3     | FR-3.3: Opportunity Qualification . . . . .         | 18        |
| 9.4       | FR-4: Automated Playbook Recommendations . . . . .  | 18        |
| 9.4.1     | FR-4.1: Playbook Library . . . . .                  | 18        |
| 9.4.2     | FR-4.2: Playbook Matching Engine . . . . .          | 19        |
| 9.4.3     | FR-4.3: Personalized Messaging . . . . .            | 19        |
| 9.5       | FR-5: Outreach Cadence Optimization . . . . .       | 19        |
| 9.5.1     | FR-5.1: Multi-Channel Sequencing . . . . .          | 19        |
| 9.5.2     | FR-5.2: Task Automation . . . . .                   | 19        |
| 9.6       | FR-6: Sentiment & Engagement Analysis . . . . .     | 20        |
| 9.6.1     | FR-6.1: Email Sentiment Tracking . . . . .          | 20        |
| 9.6.2     | FR-6.2: Engagement Monitoring . . . . .             | 20        |
| 9.7       | FR-7: Analytics & Reporting . . . . .               | 20        |
| 9.7.1     | FR-7.1: Renewal Pipeline Dashboard . . . . .        | 20        |
| 9.7.2     | FR-7.2: Expansion Revenue Tracking . . . . .        | 20        |
| 9.7.3     | FR-7.3: Agent Performance Metrics . . . . .         | 20        |
| 9.7.4     | FR-7.4: Executive Reports . . . . .                 | 21        |
| 9.8       | FR-8: Integration Requirements . . . . .            | 21        |
| 9.8.1     | FR-8.1: Salesforce Integration . . . . .            | 21        |
| 9.8.2     | FR-8.2: Billing Platform Integration . . . . .      | 21        |
| 9.8.3     | FR-8.3: Analytics Platform Integration . . . . .    | 21        |
| 9.9       | FR-9: Notification System . . . . .                 | 21        |
| 9.9.1     | FR-9.1: Alert Delivery . . . . .                    | 21        |
| 9.9.2     | FR-9.2: Alert Types . . . . .                       | 21        |
| 9.10      | FR-10: Security & Compliance . . . . .              | 22        |
| 9.10.1    | FR-10.1: Authentication & Authorization . . . . .   | 22        |
| 9.10.2    | FR-10.2: Data Privacy . . . . .                     | 22        |
| <b>10</b> | <b>Flow Diagram / Architecture Diagram</b>          | <b>22</b> |
| 10.1      | System Architecture Overview . . . . .              | 22        |
| 10.2      | Data Flow . . . . .                                 | 22        |
| 10.3      | Architecture Components . . . . .                   | 23        |
| 10.4      | Detailed Flow Descriptions . . . . .                | 24        |
| 10.4.1    | Flow 1: Churn Risk Detection . . . . .              | 24        |
| 10.4.2    | Flow 2: Upsell Opportunity Identification . . . . . | 24        |
| 10.4.3    | Flow 3: Renewal Forecast Generation . . . . .       | 24        |
| <b>11</b> | <b>Competitive Analysis</b>                         | <b>25</b> |
| 11.1      | Market Landscape . . . . .                          | 25        |
| 11.2      | Primary Competitors . . . . .                       | 25        |
| 11.2.1    | Gainsight Customer Success . . . . .                | 25        |
| 11.2.2    | ChurnZero . . . . .                                 | 25        |
| 11.2.3    | Yagna iQ Channel Ecosystem Platform . . . . .       | 26        |
| 11.2.4    | Totango . . . . .                                   | 27        |
| 11.2.5    | Catalyst (formerly ProsperWorks) . . . . .          | 27        |
| 11.3      | Innovation & Differentiation Strategy . . . . .     | 28        |
| 11.3.1    | Core Innovation: The "Agentic" Shift . . . . .      | 28        |

|                                                   |           |
|---------------------------------------------------|-----------|
| 11.3.2 Our Competitive Advantages . . . . .       | 28        |
| 11.4 Feature Comparison Matrix . . . . .          | 28        |
| 11.5 References . . . . .                         | 29        |
| <b>12 Assumptions &amp; Constraints</b>           | <b>29</b> |
| 12.1 Assumptions . . . . .                        | 29        |
| 12.2 Constraints . . . . .                        | 29        |
| <b>13 Success Metrics</b>                         | <b>30</b> |
| 13.1 Technical KPIs . . . . .                     | 30        |
| 13.2 Business KPIs . . . . .                      | 30        |
| <b>14 Risk Management</b>                         | <b>30</b> |
| 14.1 Technical Risks . . . . .                    | 30        |
| 14.2 Business Risks . . . . .                     | 30        |
| <b>15 Appendices</b>                              | <b>31</b> |
| 15.1 Appendix A: Glossary . . . . .               | 31        |
| 15.2 Appendix B: API Endpoints (Sample) . . . . . | 31        |
| 15.3 Appendix C: Contact Information . . . . .    | 31        |
| <b>16 Document Approval</b>                       | <b>31</b> |

# 1 Project Overview

## 1.1 Project Title

Renewal & Upsell Advisor - AI-Powered Revenue Protection and Expansion Agent

## 1.2 Agent Identification

- **Agent ID:** S-007
- **Agent Name:** Renewal & Upsell Advisor
- **Suite Category:** Sales Suite
- **Version:** 1.0

## 1.3 Executive Summary

The Renewal & Upsell Advisor is an intelligent AI agent designed to monitor customer contracts, usage patterns, and sentiment signals to predict renewal risk and identify upsell opportunities. The agent provides actionable recommendations including renewal plays, outreach cadences, and personalized value messaging to protect Annual Recurring Revenue (ARR) and maximize expansion revenue. By leveraging machine learning algorithms and real-time data integration from CRM, billing, and analytics platforms, the agent enables sales and customer success teams to take proactive measures before churn occurs.

# 2 Objectives

## 2.1 Primary Objectives

1. **Renewal Risk Prediction:** Implement predictive models to identify accounts at risk of churn 60-90 days before contract expiration with 85% accuracy.
2. **Upsell Opportunity Detection:** Automatically surface high-probability upsell and cross-sell opportunities based on product usage, contract value, and customer behavior patterns.
3. **Automated Playbook Execution:** Generate and execute personalized renewal and expansion playbooks with recommended outreach timing, messaging, and engagement strategies.
4. **ARR Protection:** Reduce revenue churn by 20-30% through early intervention and proactive customer engagement.
5. **Revenue Expansion:** Increase expansion revenue by 15-25% by identifying and converting qualified upsell opportunities.

## 2.2 Secondary Objectives

1. Enable real-time visibility into renewal pipeline health and revenue forecasting.
2. Reduce manual effort for sales and CS teams by automating renewal tracking and risk monitoring.
3. Provide data-driven insights to optimize pricing and packaging strategies.
4. Integrate sentiment analysis from customer communications to enhance prediction accuracy.
5. Generate automated renewal reports and executive dashboards.

## 3 Scope

### 3.1 In-Scope

- **Renewal Risk Management:**

- Contract expiration tracking and automated alerts
- Multi-factor churn prediction using ML models
- Customer health scoring based on usage, engagement, and sentiment
- Risk segmentation (High, Medium, Low) with confidence scores

- **Upsell & Cross-sell Intelligence:**

- Product adoption and feature usage analysis
- Revenue expansion opportunity identification
- Cross-sell recommendations based on customer profile and behavior
- Whitespace analysis for untapped product offerings

- **Automated Playbook Recommendations:**

- Personalized renewal and expansion plays
- Multi-channel outreach cadence suggestions (Email, Phone, Meeting)
- Value messaging templates tailored to customer pain points and outcomes
- Best-action recommendations with success probability scores

- **Integration Capabilities:**

- Primary: Salesforce CRM, Stripe/Chargebee billing platforms, Google Analytics/Mixpanel
- Secondary: Customer Success platforms (Gainsight, ChurnZero), Email (Gmail, Outlook), Product analytics tools
- Webhook-based real-time data synchronization

- **Analytics & Reporting:**

- Renewal forecast dashboard with ARR projections
- Pipeline health metrics and trend analysis
- Agent performance tracking and recommendation outcomes
- Executive summary reports with KPI visualization

### 3.2 Out-of-Scope

- Contract negotiation and legal document generation
- Direct payment processing and transaction handling
- Customer onboarding and implementation workflows
- Product roadmap planning and feature prioritization
- Marketing campaign creation and execution
- Technical support ticket management
- Full financial accounting and revenue recognition
- Integration with non-B2B SaaS business models (e.g., e-commerce, retail)
- Manual override of contract terms or pricing
- Real-time voice call monitoring and transcription

## 4 Business Scenario

### 4.1 Use Case Context

Enterprise SaaS companies typically face 5-7% annual revenue churn and miss 30-40% of expansion opportunities due to reactive renewal processes and lack of visibility into customer health signals. Sales and Customer Success teams often discover renewal risks too late to intervene effectively, and upsell opportunities are identified manually through sporadic account reviews.

### 4.2 Primary Business Scenario

**Scenario:** Proactive Renewal Risk Mitigation and Expansion Revenue Capture

**Actors:**

- Account Executives (AE)
- Customer Success Managers (CSM)
- Revenue Operations Manager
- VP of Sales

**Example Workflow:**

**Day 0-90 Before Renewal:**

1. TechCorp has 150 enterprise customers with contracts renewing in the next 90 days, representing \$8.5M in ARR.
2. The Renewal & Upsell Advisor ingests data from Salesforce (contract dates, deal history), Stripe (payment history, plan details), and analytics platforms (product usage, feature adoption).
3. The agent identifies 12 high-risk accounts (\$1.2M ARR) showing signals: declining login frequency (45% drop), increased support tickets (3x average), late payment history, and low NPS scores.

**Day 1-30 - Risk Detection & Alert:**

1. Agent sends automated Slack alert to assigned CSM: "Account XYZ Corp (\$180K ARR) - HIGH RENEWAL RISK (72% probability). Key factors: 60% usage decline, 2 unresolved support escalations, contract expires Jan 15."
2. CSM receives recommended playbook: "Executive Business Review + Value Realization Workshop" with 3 personalized email templates emphasizing ROI metrics specific to XYZ Corp's use case.
3. Agent suggests optimal outreach timing: "Schedule call within 5 days, follow-up every 7 days".

**Day 30-60 - Upsell Opportunity Detection:**

1. For 35 healthy accounts (\$2.8M ARR), agent identifies 18 expansion opportunities based on: approaching user license limits (85% utilization), high engagement with premium features, positive sentiment in recent emails.
2. AE receives notification: "ABC Company (\$250K ARR) - UPSELL OPPORTUNITY: Add 25 seats + Premium Analytics Module. Estimated expansion: \$75K. Current plan utilization: 92%".
3. Agent generates value messaging highlighting: "Based on your team's usage of Report Builder (1,250 reports/month), upgrading to Premium Analytics will unlock automated insights and save 15 hours/week".

**Day 60-90 - Renewal Execution:**

1. Agent tracks engagement metrics from outreach campaigns and updates risk scores in real-time.
2. 8 of 12 high-risk accounts move to medium risk after successful intervention.
3. Agent auto-generates renewal proposals in Salesforce with recommended contract terms.
4. Revenue Operations receives consolidated forecast: "Projected renewals: \$7.9M (93% retention), Expansion pipeline: \$850K".

**Business Outcome:**

- Churn reduced from 7% to 4% (\$255K ARR saved).



- Expansion revenue increased by 22% (\$680K captured vs. \$557K historical average).
- Sales team productivity increased by 35% through automated playbook recommendations.
- Renewal forecast accuracy improved to 92%.

## 5 Milestones & Timeline

### 5.1 7-Day Development Sprint

**Overall Duration:** 7 days (168 hours)

**Sprint Goal:** Deploy production-ready Renewal & Upsell Advisor agent with core ML models, integration pipelines, and automated playbook engine.

### 5.2 Detailed Task Breakdown

| Day     | Tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Duration |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Day 1   | <b>Architecture &amp; Setup</b> <ul style="list-style-type: none"> <li>- Environment provisioning (AWS/GCP infrastructure)</li> <li>- Database schema design (PostgreSQL/MongoDB)</li> <li>- API gateway configuration</li> <li>- Authentication &amp; authorization setup</li> <li>- CI/CD pipeline establishment</li> <li>- Development environment configuration</li> </ul>                                                                                                                                                                                                                   | 8 hours  |
| Day 1-2 | <b>Integration Layer Development</b> <ul style="list-style-type: none"> <li>- Salesforce API integration (REST/SOAP)</li> <li>- Stripe/Chargebee webhook setup</li> <li>- OAuth 2.0 authentication flows</li> <li>- Data extraction pipelines (contracts, billing, accounts)</li> <li>- Real-time event listeners</li> <li>- Error handling &amp; retry mechanisms</li> <li>- Analytics platform connectors (Segment, Mixpanel)</li> </ul>                                                                                                                                                       | 14 hours |
| Day 2-3 | <b>Data Pipeline &amp; Feature Engineering</b> <ul style="list-style-type: none"> <li>- ETL pipeline development</li> <li>- Data normalization &amp; cleansing</li> <li>- Feature extraction engine: <ul style="list-style-type: none"> <li>- Contract metadata (renewal date, ARR, tenure)</li> <li>- Usage metrics (login frequency, feature adoption)</li> <li>- Behavioral signals (support tickets, NPS)</li> <li>- Payment history (late payments, failed charges)</li> <li>- Time-series aggregation (7/30/90-day windows)</li> </ul> </li> <li>- Feature store implementation</li> </ul> | 12 hours |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Day 3-4                         | <b>ML Model Development</b> <ul style="list-style-type: none"> <li>- Churn prediction model (Gradient Boosting/Random Forest)</li> <li>- Training on historical renewal outcomes</li> <li>- Model evaluation (Precision, Recall, F1-score)</li> <li>- Hyperparameter tuning</li> <li>- Upsell propensity scoring model</li> <li>- Cross-sell recommendation engine</li> <li>- Model versioning &amp; registry setup</li> <li>- Inference API endpoints</li> </ul>                          | 16 hours         |
| Day 4-5                         | <b>Playbook Engine &amp; Recommendation System</b> <ul style="list-style-type: none"> <li>- Rule-based playbook matching logic</li> <li>- Outreach cadence optimizer</li> <li>- Value messaging template engine</li> <li>- Personalization layer (dynamic variables)</li> <li>- Recommendation scoring algorithm</li> <li>- Action prioritization framework</li> <li>- A/B testing infrastructure for playbooks</li> </ul>                                                                 | 14 hours         |
| Day 5-6                         | <b>Notification &amp; Workflow Automation</b> <ul style="list-style-type: none"> <li>- Slack/Teams webhook integration</li> <li>- Email notification service (SendGrid/AWS SES)</li> <li>- In-app alert system</li> <li>- Task automation in Salesforce (create opportunities, tasks)</li> <li>- Scheduled batch jobs (daily risk scoring)</li> <li>- Event-driven trigger system</li> <li>- Workflow orchestration (Apache Airflow)</li> </ul>                                            | 12 hours         |
| Day 6                           | <b>Dashboard &amp; Analytics UI</b> <ul style="list-style-type: none"> <li>- Renewal pipeline dashboard (React/Vue.js)</li> <li>- Risk heatmap visualization</li> <li>- ARR forecasting charts</li> <li>- Agent performance metrics</li> <li>- Drill-down account details view</li> <li>- Export functionality (CSV, PDF)</li> <li>- Responsive design implementation</li> </ul>                                                                                                           | 10 hours         |
| Day 7                           | <b>Testing, Deployment &amp; Documentation</b> <ul style="list-style-type: none"> <li>- Unit testing (Jest, PyTest)</li> <li>- Integration testing (API contract validation)</li> <li>- End-to-end scenario testing</li> <li>- Performance testing (load, stress)</li> <li>- Security testing (penetration, vulnerability scan)</li> <li>- Production deployment</li> <li>- Rollback procedures</li> <li>- Technical documentation</li> <li>- User guide &amp; playbook library</li> </ul> | 14 hours         |
| <b>Total Development Hours:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>100 hours</b> |

### 5.3 Critical Path Dependencies

1. Integration Layer must complete before Data Pipeline development
2. Feature Engineering required before ML Model training
3. ML Models must be operational before Playbook Engine development
4. All backend services completed before Dashboard UI build

## 6 Roles & Responsibilities

### 6.1 Technical Team Structure

| Role                               | Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tech Lead / Architect              | <ul style="list-style-type: none"> <li>- System architecture design &amp; technical decision-making</li> <li>- Technology stack selection</li> <li>- Integration strategy definition</li> <li>- Code review &amp; quality assurance</li> <li>- Team coordination &amp; sprint planning</li> <li>- Risk identification &amp; mitigation</li> <li>- Performance optimization &amp; scalability planning</li> </ul>                                                                 |
| Backend Engineer 1 (Integrations)  | <ul style="list-style-type: none"> <li>- Salesforce API integration development</li> <li>- Stripe/Chargebee webhook implementation</li> <li>- OAuth authentication flows</li> <li>- REST API endpoint development</li> <li>- Data synchronization pipelines</li> <li>- Error handling &amp; logging</li> <li>- API documentation (Swagger/OpenAPI)</li> </ul>                                                                                                                    |
| Backend Engineer 2 (Data Pipeline) | <ul style="list-style-type: none"> <li>- ETL pipeline architecture &amp; implementation</li> <li>- Data transformation &amp; normalization logic</li> <li>- Feature engineering workflows</li> <li>- Database schema design &amp; optimization</li> <li>- Batch processing jobs (Apache Airflow)</li> <li>- Data quality validation &amp; monitoring</li> <li>- Time-series data aggregation</li> </ul>                                                                          |
| ML Engineer                        | <ul style="list-style-type: none"> <li>- ML model development (churn prediction, upsell scoring)</li> <li>- Feature selection &amp; dimensionality reduction</li> <li>- Model training, validation &amp; hyperparameter tuning</li> <li>- ML pipeline orchestration (MLflow, Kubeflow)</li> <li>- Model deployment &amp; serving infrastructure</li> <li>- A/B testing framework for model versions</li> <li>- Model performance monitoring &amp; retraining triggers</li> </ul> |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full-Stack Developer        | <ul style="list-style-type: none"> <li>- Frontend dashboard development (React/Vue.js)</li> <li>- Backend API integration for UI</li> <li>- Data visualization components (Chart.js, D3.js)</li> <li>- Responsive design implementation</li> <li>- User authentication &amp; session management</li> <li>- Playbook recommendation UI</li> <li>- Export functionality (reports, CSV)</li> </ul>                                                                                                                                           |
| DevOps Engineer             | <ul style="list-style-type: none"> <li>- Cloud infrastructure setup (AWS/GCP/Azure)</li> <li>- Kubernetes cluster provisioning &amp; management</li> <li>- CI/CD pipeline configuration (Jenkins, GitHub Actions)</li> <li>- Container orchestration (Docker, Kubernetes)</li> <li>- Monitoring &amp; alerting setup (Prometheus, Grafana)</li> <li>- Secrets management (Vault, AWS Secrets Manager)</li> <li>- Production deployment &amp; rollback procedures</li> <li>- Infrastructure-as-Code (Terraform, CloudFormation)</li> </ul> |
| QA Engineer                 | <ul style="list-style-type: none"> <li>- Test plan creation &amp; execution</li> <li>- Unit test development (Jest, PyTest)</li> <li>- Integration testing (API contracts, data flows)</li> <li>- End-to-end scenario testing (Selenium, Cypress)</li> <li>- Performance &amp; load testing (JMeter, Locust)</li> <li>- Security testing coordination</li> <li>- Bug tracking &amp; regression testing</li> <li>- Test automation framework setup</li> </ul>                                                                              |
| Data Scientist (Consultant) | <ul style="list-style-type: none"> <li>- Churn prediction model design &amp; algorithm selection</li> <li>- Statistical analysis of renewal patterns</li> <li>- Feature importance analysis</li> <li>- Model interpretability (SHAP, LIME)</li> <li>- Business metrics validation</li> <li>- Recommendation logic optimization</li> <li>- Training data labeling strategy</li> </ul>                                                                                                                                                      |

## 6.2 Team Size

- **Core Team:** 7 members (Tech Lead, 2 Backend Engineers, 1 ML Engineer, 1 Full-Stack Developer, 1 DevOps Engineer, 1 QA Engineer)
- **Extended Support:** 1 Data Scientist (part-time consultant)

## 7 Deliverables

### 7.1 Technical Deliverables

#### 1. Renewal & Upsell Advisor Agent - Production Application

- Dockerized microservices architecture
- Deployed on cloud infrastructure (AWS/GCP)
- Auto-scaling configuration

- 99.5% uptime SLA

## 2. Machine Learning Models

- Churn prediction model (Gradient Boosting Classifier)
- Upsell propensity scoring model
- Cross-sell recommendation engine
- Model artifacts & versioning (MLflow registry)
- Inference API endpoints

## 3. Integration Connectors

- Salesforce bidirectional sync
- Stripe/Chargebee webhook listeners
- Google Analytics / Mixpanel event streaming
- Customer Success platform connectors (optional: Gainsight, ChurnZero)
- Email service integration (Gmail, Outlook via MS Graph API)

## 4. Automated Playbook Engine

- 15+ pre-built renewal & upsell playbooks
- Personalized messaging template library
- Outreach cadence optimizer
- Multi-channel recommendation system

## 5. Analytics Dashboard

- Renewal pipeline overview
- Risk heatmap & account segmentation
- ARR forecast visualization
- Upsell opportunity tracker
- Agent performance metrics
- Exportable reports (PDF, CSV)

## 6. Notification System

- Slack/Teams integration
- Email alerts (SendGrid/AWS SES)
- In-app notification center
- Configurable alert rules

## 7. API Documentation

- REST API specification (OpenAPI 3.0)
- Authentication guide
- Webhook payload schemas
- Code examples (Python, JavaScript)

## 7.2 Documentation Deliverables

### 1. Technical Documentation

- System architecture diagram
- Database schema documentation
- API reference guide
- Deployment runbook
- Disaster recovery procedures

### 2. User Documentation

- User manual (Sales & CS teams)
- Playbook customization guide
- Dashboard navigation tutorial
- Integration setup guide

### 3. Training Materials

- Video walkthrough (15 minutes)
- Quick start guide
- FAQ document
- Best practices for renewal management

## 8 Tech Stack

### 8.1 Frontend Technologies

- **Framework:** React 18.x or Vue.js 3.x
- **State Management:** Redux Toolkit / Vuex
- **UI Library:** Material-UI / Ant Design
- **Data Visualization:** Chart.js, Recharts, D3.js
- **HTTP Client:** Axios
- **Build Tool:** Vite / Webpack 5
- **Testing:** Jest, React Testing Library, Cypress

## 8.2 Backend Technologies

- **Primary Language:** Python 3.11+
- **Web Framework:** FastAPI / Django REST Framework
- **Task Queue:** Celery with Redis broker
- **Workflow Orchestration:** Apache Airflow
- **Async Processing:** asyncio, aiohttp
- **API Standards:** REST, GraphQL (optional)
- **Authentication:** OAuth 2.0, JWT
- **API Documentation:** Swagger/OpenAPI 3.0

## 8.3 Machine Learning Stack

- **ML Framework:** Scikit-learn, XGBoost, LightGBM
- **Deep Learning:** TensorFlow / PyTorch (optional for advanced NLP)
- **NLP:** spaCy, Hugging Face Transformers (sentiment analysis)
- **Feature Store:** Feast or custom implementation
- **Model Management:** MLflow
- **Model Serving:** TensorFlow Serving / TorchServe
- **Experimentation:** Weights & Biases (W&B)

## 8.4 Database & Storage

- **Primary Database:** PostgreSQL 15+ (relational data)
- **NoSQL Database:** MongoDB / DynamoDB (event logs, unstructured data)
- **Caching Layer:** Redis 7+ (session, API response cache)
- **Data Warehouse:** Snowflake / Google BigQuery (analytics)
- **Object Storage:** AWS S3 / Google Cloud Storage (model artifacts, reports)
- **Vector Database:** Pinecone / Weaviate (for similarity search, optional)

## 8.5 Integration & APIs

- **Salesforce:** REST API, SOAP API, Bulk API, Streaming API
- **Stripe:** REST API, Webhooks
- **Chargebee:** REST API, Webhooks
- **Google Analytics:** Measurement Protocol, Reporting API
- **Mixpanel:** Export API, Engage API
- **Slack:** Web API, Webhook
- **Microsoft Teams:** Graph API, Incoming Webhooks
- **Email:** SendGrid, AWS SES, Gmail API, MS Graph API
- **Customer Success Platforms:** Gainsight API, ChurnZero API

## 8.6 DevOps & Infrastructure

- **Cloud Platform:** AWS / Google Cloud Platform / Azure
- **Containerization:** Docker
- **Orchestration:** Kubernetes (EKS, GKE, AKS)
- **CI/CD:** GitHub Actions / Jenkins / GitLab CI
- **Infrastructure-as-Code:** Terraform / AWS CloudFormation
- **Monitoring:** Prometheus, Grafana, Datadog
- **Logging:** ELK Stack (Elasticsearch, Logstash, Kibana) / Splunk
- **Error Tracking:** Sentry
- **Secrets Management:** AWS Secrets Manager / HashiCorp Vault
- **Load Balancer:** AWS ALB / NGINX

## 8.7 Development Tools

- **Version Control:** Git, GitHub/GitLab
- **IDE:** VS Code, PyCharm, Jupyter Notebook
- **API Testing:** Postman, Insomnia
- **Collaboration:** Jira, Confluence, Slack
- **Code Quality:** SonarQube, ESLint, Pylint, Black (formatter)



## 9 Functional Requirements

### 9.1 FR-1: Contract & Renewal Tracking

#### 9.1.1 FR-1.1: Contract Data Ingestion

- System SHALL ingest contract data from Salesforce including: contract ID, account name, ARR/MRR, start date, end date, renewal date, contract terms
- System SHALL automatically sync contract updates within 5 minutes of changes in source system
- System SHALL support bulk import of historical contract data (minimum 5 years)

#### 9.1.2 FR-1.2: Renewal Timeline Monitoring

- System SHALL track all contracts with renewal dates within 90-day window
- System SHALL automatically create renewal opportunity records in Salesforce 90 days before expiration
- System SHALL categorize renewals by time horizon: Immediate (0-30 days), Near-term (31-60 days), Future (61-90 days)

### 9.2 FR-2: Churn Risk Prediction

#### 9.2.1 FR-2.1: Risk Scoring Model

- System SHALL calculate renewal risk score (0-100) for each account using ML model
- Risk score SHALL incorporate features: product usage (login frequency, feature adoption), support metrics (ticket volume, resolution time), payment history (late payments, failed charges), engagement (email opens, meeting attendance), sentiment (NPS, survey responses)
- System SHALL update risk scores daily via batch processing
- System SHALL provide 85% prediction accuracy (validated on historical data)

#### 9.2.2 FR-2.2: Risk Segmentation

- System SHALL classify accounts into risk categories:
  - High Risk: Score 70-100 (churn probability >50%)
  - Medium Risk: Score 40-69 (churn probability 25-50%)
  - Low Risk: Score 0-39 (churn probability <25%)
- System SHALL display top 5 risk factors for each account
- System SHALL provide confidence intervals for predictions

### 9.2.3 FR-2.3: Early Warning Alerts

- System SHALL send real-time alerts when account moves to High Risk category
- Alerts SHALL include: account name, ARR at risk, risk score, primary risk factors, recommended actions
- System SHALL support alert routing rules (assign to CSM, AE, or team lead based on account ownership)

## 9.3 FR-3: Upsell & Cross-sell Identification

### 9.3.1 FR-3.1: Expansion Opportunity Detection

- System SHALL identify upsell opportunities based on:
  - User license utilization (threshold: >80% of purchased seats)
  - Feature usage patterns (high engagement with premium features on basic plan)
  - API usage limits (approaching rate limits)
  - Storage consumption (exceeding 75% of allocated quota)
- System SHALL calculate expansion revenue potential for each opportunity
- System SHALL rank opportunities by expected value and win probability

### 9.3.2 FR-3.2: Cross-sell Recommendations

- System SHALL recommend complementary products based on:
  - Current product usage (collaborative filtering)
  - Industry segment and company size
  - Similar customer adoption patterns
- System SHALL support whitespace analysis (identify unused products in customer's plan tier)

### 9.3.3 FR-3.3: Opportunity Qualification

- System SHALL assign propensity score (0-100) to each upsell opportunity
- System SHALL filter opportunities with score >60 for proactive outreach
- System SHALL estimate deal size and expected close date

## 9.4 FR-4: Automated Playbook Recommendations

### 9.4.1 FR-4.1: Playbook Library

- System SHALL include minimum 15 pre-built playbooks:
  - Renewal Playbooks: High-Risk Intervention, Executive Business Review, Value Realization Workshop, Competitive Displacement Defense

- Upsell Playbooks: License Expansion, Feature Upgrade, Multi-Year Commitment, Premium Support Add-on
- Each playbook SHALL contain: trigger conditions, outreach cadence (timeline), email/call script templates, success metrics

#### **9.4.2 FR-4.2: Playbook Matching Engine**

- System SHALL automatically recommend playbook based on: account risk level, contract value, industry vertical, customer lifecycle stage
- System SHALL support A/B testing of playbook effectiveness
- System SHALL allow CSM/AE to override recommendations

#### **9.4.3 FR-4.3: Personalized Messaging**

- System SHALL generate personalized email templates with dynamic variables:
  - Customer-specific ROI metrics
  - Recent usage statistics
  - Relevant case studies from similar companies
  - Competitive insights (if applicable)
- Templates SHALL maintain brand voice and compliance requirements

### **9.5 FR-5: Outreach Cadence Optimization**

#### **9.5.1 FR-5.1: Multi-Channel Sequencing**

- System SHALL recommend outreach sequence across channels: Email, Phone Call, LinkedIn Message, In-Product Notification
- System SHALL optimize timing based on: historical engagement patterns, recipient time zone, industry benchmarks
- System SHALL support customizable cadence templates (3-touch, 5-touch, 7-touch sequences)

#### **9.5.2 FR-5.2: Task Automation**

- System SHALL automatically create tasks in Salesforce for recommended actions
- System SHALL set due dates aligned with playbook timeline
- System SHALL send reminders 24 hours before task due date

## 9.6 FR-6: Sentiment & Engagement Analysis

### 9.6.1 FR-6.1: Email Sentiment Tracking

- System SHALL analyze sentiment from customer emails using NLP (spaCy, Hugging Face)
- System SHALL classify sentiment as Positive, Neutral, Negative with confidence score
- System SHALL detect sentiment shifts (declining trend over 30-day window)

### 9.6.2 FR-6.2: Engagement Monitoring

- System SHALL track engagement metrics: email open rate, link click rate, meeting acceptance rate, response time
- System SHALL flag disengaged accounts (no response to 3+ outreach attempts)

## 9.7 FR-7: Analytics & Reporting

### 9.7.1 FR-7.1: Renewal Pipeline Dashboard

- Dashboard SHALL display:
  - Total ARR at risk (segmented by risk level)
  - Renewal rate forecast (current quarter, next quarter)
  - Top 10 at-risk accounts by ARR
  - Churn reason analysis (historical)
- Dashboard SHALL support filtering by: renewal date range, account segment, assigned CSM/AE

### 9.7.2 FR-7.2: Expansion Revenue Tracking

- Dashboard SHALL display:
  - Total expansion pipeline value
  - Upsell conversion rate
  - Average deal size by opportunity type
  - Time-to-close for expansion deals

### 9.7.3 FR-7.3: Agent Performance Metrics

- System SHALL track:
  - Prediction accuracy (renewal outcome vs. predicted risk)
  - Playbook effectiveness (conversion rate by playbook type)
  - Alert response time (time from alert to first action)
  - Revenue saved via early intervention

#### **9.7.4 FR-7.4: Executive Reports**

- System SHALL generate weekly executive summary including: ARR retention forecast, Top risks & mitigations, Expansion opportunities closed, YoY performance trends
- Reports SHALL be exportable in PDF and CSV formats

### **9.8 FR-8: Integration Requirements**

#### **9.8.1 FR-8.1: Salesforce Integration**

- System SHALL sync bidirectionally: Accounts, Opportunities, Contracts, Tasks, Activities
- System SHALL support custom Salesforce objects
- System SHALL handle API rate limits (5,000 calls/day for Enterprise edition)

#### **9.8.2 FR-8.2: Billing Platform Integration**

- System SHALL ingest data from Stripe/Chargebee: Subscription details, Payment history, Plan changes, Failed charges
- System SHALL process webhook events in real-time

#### **9.8.3 FR-8.3: Analytics Platform Integration**

- System SHALL collect product usage data from: Google Analytics (page views, session duration), Mixpanel (event tracking, funnel analysis), Custom analytics APIs

### **9.9 FR-9: Notification System**

#### **9.9.1 FR-9.1: Alert Delivery**

- System SHALL support notification channels: Slack (direct message, channel post), Email (digest or real-time), In-app notification center, Microsoft Teams
- Users SHALL configure notification preferences per alert type

#### **9.9.2 FR-9.2: Alert Types**

- System SHALL generate alerts for:
  - High-risk account detection
  - Upsell opportunity identification
  - Contract expiration (30/60/90-day reminders)
  - Failed payment events
  - Significant usage decline

## 9.10 FR-10: Security & Compliance

### 9.10.1 FR-10.1: Authentication & Authorization

- System SHALL support SSO via OAuth 2.0 (Okta, Azure AD)
- System SHALL enforce role-based access control (RBAC): Admin, Sales Manager, CSM, AE, Read-Only
- System SHALL log all user actions for audit trail

### 9.10.2 FR-10.2: Data Privacy

- System SHALL encrypt data at rest (AES-256) and in transit (TLS 1.3)
- System SHALL support GDPR/CCPA compliance (data anonymization, deletion)
- System SHALL mask sensitive PII in non-production environments

## 10 Flow Diagram / Architecture Diagram

### 10.1 System Architecture Overview

The Renewal & Upsell Advisor follows a microservices architecture with three primary layers:

1. **Integration Layer:** Handles bidirectional data synchronization with external systems (Salesforce, Stripe, Analytics platforms)
2. **Processing Layer:** Executes data transformation, feature engineering, ML inference, and business logic
3. **Presentation Layer:** Provides user interfaces (dashboard) and notification delivery

### 10.2 Data Flow

#### Ingestion Phase:

1. External systems (Salesforce, Stripe, Analytics) push data via webhooks or API polling
2. Integration Service validates and queues incoming data
3. ETL Pipeline processes raw data and stores in Data Warehouse

#### Processing Phase:

1. Feature Engineering Service extracts signals from historical data
2. ML Inference Service calculates risk scores and upsell propensity
3. Playbook Engine matches accounts to recommended plays

4. Results stored in PostgreSQL for real-time access

**Action Phase:**

1. Notification Service triggers alerts (Slack, Email) based on threshold rules
2. Workflow Automation creates tasks/opportunities in Salesforce
3. Dashboard displays insights and recommendations to users

## 10.3 Architecture Components

**Microservices:**

- Integration Service (Node.js / Python FastAPI)
- ETL Pipeline Service (Apache Airflow)
- Feature Engineering Service (Python)
- ML Inference Service (Python + MLflow)
- Playbook Recommendation Engine (Python)
- Notification Service (Python + Celery)
- Dashboard API (FastAPI)
- Frontend Application (React)

**Data Stores:**

- PostgreSQL: Contracts, accounts, opportunities, tasks
- MongoDB: Event logs, unstructured customer interactions
- Redis: Session cache, API response cache, task queue
- Snowflake/BigQuery: Historical analytics, reporting
- S3: Model artifacts, exported reports

**Supporting Infrastructure:**

- API Gateway (Kong / AWS API Gateway): Rate limiting, authentication
- Message Queue (RabbitMQ / AWS SQS): Async task processing
- Load Balancer (AWS ALB): Traffic distribution
- Monitoring Stack (Prometheus + Grafana): Metrics, alerts

## 10.4 Detailed Flow Descriptions

### 10.4.1 Flow 1: Churn Risk Detection

1. Daily batch job retrieves contracts expiring in 90 days from Salesforce
2. System fetches associated usage data (Google Analytics), payment history (Stripe), support tickets (Zendesk)
3. Feature Engineering Service calculates: 30-day login frequency decline, feature adoption rate, NPS score, payment punctuality score
4. ML Model predicts churn probability for each account
5. Accounts with risk score  $>70$  trigger High Risk alert
6. Playbook Engine recommends intervention playbook
7. Notification sent to assigned CSM via Slack with action items
8. System creates task in Salesforce: "Schedule Executive Business Review"

### 10.4.2 Flow 2: Upsell Opportunity Identification

1. Real-time webhook from product analytics detects: Account XYZ reaches 85% user license utilization
2. System retrieves account profile: current plan tier, contract value, renewal date
3. Upsell Propensity Model calculates expansion likelihood (78% probability)
4. System estimates expansion value: +25 seats = \$75K additional ARR
5. Opportunity automatically created in Salesforce with "Qualified" stage
6. Personalized email template generated: "Your team is growing fast. Unlock 25 more seats with preferred pricing."
7. AE receives in-app notification with recommended talk track
8. System schedules 3-touch follow-up sequence (Email Day 1, Call Day 3, Email Day 7)

### 10.4.3 Flow 3: Renewal Forecast Generation

1. Monthly batch job aggregates all renewals for next quarter
2. System calculates weighted renewal forecast:  $(\text{Low Risk ARR} \times 95\%) + (\text{Medium Risk} \times 75\%) + (\text{High Risk} \times 40\%)$
3. Expansion pipeline added to base renewal forecast
4. Historical conversion rates applied to upsell opportunities
5. Dashboard displays: Total forecasted ARR, Confidence intervals (P10, P50, P90), Risk-adjusted retention rate



## 6. Executive report exported to PDF and emailed to VP Sales

**Note:** Detailed architecture diagram is provided separately as a visual chart. Reference the flow diagram PDF attached with this SRS document.

# 11 Competitive Analysis

## 11.1 Market Landscape

The renewal and upsell intelligence market includes specialized customer success platforms, revenue operations tools, and AI-powered sales enablement solutions. Key competitors offer varying degrees of automation, predictive analytics, and CRM integration.

## 11.2 Primary Competitors

### 11.2.1 Gainsight Customer Success

**Product URL:** <https://www.gainsight.com/>

**Strengths:**

- Industry-leading customer health scoring with ML-powered predictions
- Native Salesforce integration with deep data synchronization
- Comprehensive playbook automation and journey orchestration
- Strong analytics and executive reporting dashboards
- Extensive third-party integrations (200+ apps)

**Weaknesses:**

- Complex setup requiring significant admin resources
- Higher cost (enterprise pricing starts at \$50K+/year)
- Steeper learning curve for end users
- Limited flexibility for mid-market companies

**Technical Approach:** Gainsight uses Bayesian recursive models for churn prediction, combining usage telemetry, support data, and sentiment analysis. Their "Rules Engine" triggers automated workflows based on multi-dimensional health scores. Data aggregation occurs in their proprietary data warehouse with scheduled ETL jobs from source systems.

### 11.2.2 ChurnZero

**Product URL:** <https://churnzero.com/>

**Strengths:**

- Real-time customer engagement tracking and alerts
- Easy-to-use interface with faster time-to-value

- Strong renewal forecasting with Forecast Hub
- AI agents (Vibes, Harbinger, Beacon) for risk and opportunity detection
- Competitive pricing for mid-market segment

**Weaknesses:**

- Less robust analytics compared to Gainsight
- Limited customization for complex enterprise workflows
- Fewer native integrations (relies on Zapier for some connections)
- Reporting can feel rigid per user feedback

**Technical Approach:** ChurnZero emphasizes real-time event streaming from product analytics. Their AI agents use sentiment shift detection (NLP on emails/chat) and behavioral pattern matching. Renewal forecasting combines health scores with historical win/loss rates. They employ a segment-based approach for playbook recommendations.

### 11.2.3 Yagna iQ Channel Ecosystem Platform

**Product URL:** <https://www.yagnaiq.com/>

**Strengths:**

- **High ROI & Efficiency:** Proven to save 125+ hours per 100 contracts and increase on-time renewal rates by 20–25%.
- **Zero-Touch Automation:** "Set and forget" workflows for low-value, high-volume renewals, triggering proactive quotes 90/60/30 days in advance.
- **Unified Channel View:** Solves the "1000 Portal Problem" by aggregating multi-vendor data into a single dashboard for distributors and resellers.
- **Security & Integration:** ISO 27001 certified with native connectors for Salesforce, Microsoft Dynamics, and major ERPs.

**Weaknesses:**

- Highly specialized for *indirect* channel sales (Vendors → Distributors → Resellers), which may be overkill for direct-only SaaS businesses.
- UI/UX is function-first (industrial strength) rather than "consumer-grade" aesthetic compared to modern PLG tools.

**Technical Approach:** Yagna iQ operates as a cloud-native "Channel Ecosystem" platform. It ingests "Golden Record" data from vendor ERPs/CRMs and normalizes it for downstream partners. Its core engine uses AI/ML to identify "Smart Leads" for cross-selling (XSUS Cloud) and automates the entire quote-to-cash cycle for renewals (Renewal Cloud) using event-driven triggers.

#### 11.2.4 Totango

**Product URL:** <https://www.totango.com/>

**Strengths:**

- Product adoption-focused with deep usage analytics
- Flexible pricing and modular features (scale as needed)
- Strong customer segmentation capabilities
- Pre-built success plans and templates
- Good balance of simplicity and functionality

**Weaknesses:**

- Churn prediction less sophisticated than Gainsight
- Limited upsell recommendation intelligence
- Fewer automated playbook options
- Smaller user community and ecosystem

**Technical Approach:** Totango's core philosophy centers on product usage as the primary health indicator. They auto-generate health scores based on feature adoption patterns without requiring manual configuration. Segments are dynamically updated based on lifecycle stage and account attributes. Renewal risk is inferred from usage decline trends.

#### 11.2.5 Catalyst (formerly ProsperWorks)

**Product URL:** <https://www.catalyst.io/>

**Strengths:**

- Unified customer data platform (CDP) approach
- Strong revenue forecasting and pipeline management
- AI-driven insights with "moments" (trigger events)
- Modern UI with intuitive UX

**Weaknesses:**

- Relatively new player (smaller customer base)
- Fewer proven ML models compared to Gainsight
- Limited third-party integration ecosystem
- Still building out playbook automation

**Technical Approach:** Catalyst aggregates customer data into a unified profile (CDP model). Their "moments" feature detects trigger events (e.g., usage spike, support escalation) using rule-based and ML hybrid models. Revenue forecasting uses cohort analysis and time-series prediction.

### 11.3 Innovation & Differentiation Strategy

#### 11.3.1 Core Innovation: The "Agentic" Shift

The primary innovation of our Renewal & Upsell Advisor is the fundamental shift from **Passive Analytics** (Dashboards) to **Active Agency** (Co-Pilots).

- 1. From "Insights" to "Actions":** Legacy platforms (Gainsight, Yagna iQ) identify a risk and populate a dashboard, requiring a human Key Account Manager (KAM) to log in, interpretation the data, and decide on an action. **Our Innovation:** The Agent identifies the risk and *immediately drafts the remedial action* (e.g., a personalized email, a meeting agenda, or a discount offer) for the KAM to approve. The KAM manages *decisions*, not data.
- 2. Generative vs. Static Playbooks:** Competitors use rigid "If-Then" rule engines (e.g., "If health score < 50, send Email Template B"). **Our Innovation:** We utilize Large Language Models (LLMs) to generate *context-aware* playbooks on the fly. The Agent reads the last 10 emails, analyzes the support ticket tone, and generates a truly personalized message that acknowledges specific client pain points, rather than using a generic template.
- 3. Unified "Growth Loop" Architecture:** Most platforms treat "Renewals" (Defense) and "Upsells" (Offense) as separate pipelines. **Our Innovation:** We model them as a single **Revenue Optimization Problem**. Every interaction is scored for both Retention Probability and Expansion Propensity simultaneously. The Agent will suppress an upsell offer if the renewal risk is high, or leverage a successful renewal to immediately pitch an upsell, creating a continuous growth loop.

#### 11.3.2 Our Competitive Advantages

- 1. Zero-Login Experience:** The Agent lives where the KAM works (Slack, Teams, Email). There is no "platform" to learn or log into.
- 2. Hyper-Personalization at Scale:** Using Generative AI to craft unique account plans for thousands of long-tail customers that would otherwise receive generic automation.
- 3. Faster Time-to-Value:** 7-day deployment via API overlay, compared to 6–12 week implementation cycles for heavy platforms like Yagna or Gainsight.

### 11.4 Feature Comparison Matrix

| Feature           | Our Agent        | Gainsight   | ChurnZero   | Totango     | Catalyst    | Yagna iQ       |
|-------------------|------------------|-------------|-------------|-------------|-------------|----------------|
| Churn Prediction  | Action-Oriented  | Descriptive | Descriptive | Basic       | Descriptive | Analytical     |
| Upsell Propensity | Generative Pitch | Partial     | Yes         | Limited     | Yes         | Smart Leads    |
| Playbook Engine   | GenAI (Dynamic)  | Rigid Rules | Rigid Rules | Rigid Rules | Rigid Rules | Rigid Workflow |

|                  |                              |               |               |            |               |               |
|------------------|------------------------------|---------------|---------------|------------|---------------|---------------|
| Real-Time Alerts | <b>Drafts Re-sponse</b>      | Notification  | Notification  | No         | Notification  | Quote Trigger |
| User Interface   | <b>Chat/Slack (Headless)</b> | Dashboard     | Dashboard     | Dashboard  | Dashboard     | Portal        |
| Salesforce Sync  | <b>bidirectional Agent</b>   | Native Sync   | Sync          | Sync       | Native Sync   | Native Sync   |
| Deployment       | <b>7 Days (Overlay)</b>      | 8–12 weeks    | 4–6 weeks     | 2–4 weeks  | 3–5 weeks     | 4–8 weeks     |
| Pricing          | <b>Usage/Outcome</b>         | Seat-Based    | Seat-Based    | Seat-Based | Seat-Based    | Volume-Based  |
| Core AI Tech     | <b>LLM + RAG</b>             | Predictive ML | Predictive ML | Basic ML   | Predictive ML | Predictive ML |

## 11.5 References

1. Gainsight Customer Success Platform: <https://www.gainsight.com/customer-success/>
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6. ML-Based Renewal Prediction: <https://www.sirion.ai/library/contract-insights/predicting-renewal-pr>
7. Stripe + Salesforce Integration Guide: <https://candyboxcrm.com/blog/how-to-set-up-the-stripe-to-sales>
8. ARR Forecasting Methods: <https://www.drivetrain.ai/post/arr-forecasting-corkscrew-model>
9. Churn Prediction Tools 2025: <https://www.zendesk.com/in/service/customer-experience/churn-prediction-software/>
10. Sentiment Analysis for Churn: <https://insight7.io/how-to-use-sentiment-analysis-to-predict-churn/>

## 12 Assumptions & Constraints

### 12.1 Assumptions

1. Client has active Salesforce instance (Enterprise or Unlimited edition).
2. Historical data available for minimum 2 years (contracts, renewals, usage).
3. API access credentials provided for all integrated systems.
4. Dedicated technical point-of-contact available during development.
5. Cloud infrastructure budget allocated (AWS/GCP).

### 12.2 Constraints

1. 7-day development timeline (no scope creep).
2. API rate limits from external systems (Salesforce: 5,000 calls/day).
3. Data privacy regulations (GDPR, CCPA compliance required).
4. Limited budget for third-party ML services (prefer open-source).
5. Single deployment environment (production only, no separate staging initially).

## 13 Success Metrics

### 13.1 Technical KPIs

- **ML Model Accuracy:** >85% precision and recall on churn prediction
- **System Uptime:** 99.5% availability
- **API Response Time:** <200ms (P95)
- **Data Sync Latency:** <5 minutes for real-time updates
- **Alert Delivery Time:** <60 seconds from trigger event

### 13.2 Business KPIs

- **Churn Reduction:** 20-30% decrease in revenue churn
- **Expansion Revenue:** 15-25% increase in upsell conversions
- **Forecast Accuracy:** >90% accuracy on quarterly renewal predictions
- **Team Productivity:** 35% reduction in manual renewal tracking time
- **Early Intervention Rate:** 80% of high-risk accounts engaged within 5 days of alert

## 14 Risk Management

### 14.1 Technical Risks

- **Risk:** API rate limit exceeded during peak usage
  - **Mitigation:** Implement request queuing, caching, and batch processing
- **Risk:** ML model drift (accuracy degrades over time)
  - **Mitigation:** Automated weekly model performance monitoring, quarterly retraining
- **Risk:** Integration failure with external systems
  - **Mitigation:** Implement retry logic, fallback mechanisms, comprehensive error logging

### 14.2 Business Risks

- **Risk:** User adoption challenges (sales team ignores recommendations)
  - **Mitigation:** Comprehensive training, in-app guidance, demonstrate early wins
- **Risk:** Data quality issues (incomplete or inaccurate source data)
  - **Mitigation:** Data validation rules, quality scoring, alert admins to data gaps

## 15 Appendices

### 15.1 Appendix A: Glossary

- **ARR:** Annual Recurring Revenue
- **MRR:** Monthly Recurring Revenue
- **Churn:** Loss of customers or revenue
- **NPS:** Net Promoter Score
- **CSM:** Customer Success Manager
- **AE:** Account Executive
- **CRM:** Customer Relationship Management
- **ETL:** Extract, Transform, Load
- **ML:** Machine Learning
- **NLP:** Natural Language Processing

### 15.2 Appendix B: API Endpoints (Sample)

POST /api/v1/predict/churn  
GET /api/v1/renewals?risk\_level=high  
POST /api/v1/playbooks/recommend  
GET /api/v1/opportunities/upsell  
POST /api/v1/alerts/configure

### 15.3 Appendix C: Contact Information

- **Project Manager:** [Name], [Email]
- **Tech Lead:** [Name], [Email]
- **Product Owner:** [Name], [Email]

## 16 Document Approval

|                     |                 |
|---------------------|-----------------|
| <b>Prepared By:</b> | [Name], [Title] |
| <b>Date:</b>        |                 |
| <b>Reviewed By:</b> | [Name], [Title] |
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